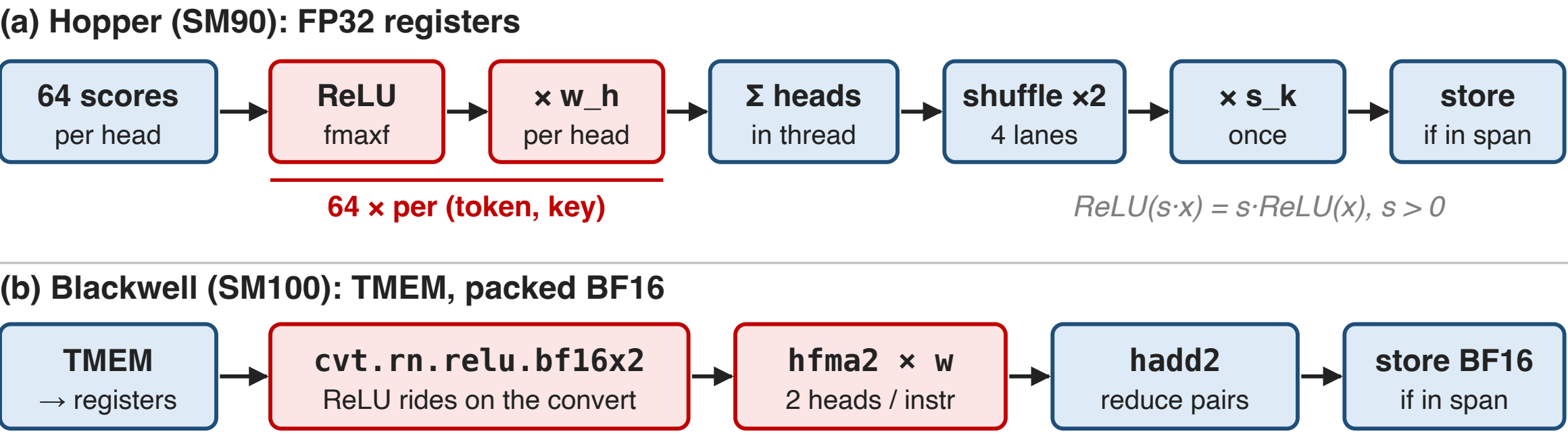

(a) Hopper (SM90): FP32 registers
64 scores
per head
ReLU
fmaxf
× w_h
per head
Σ heads
in thread
shuffle ×2
4 lanes
× s_k
once
store
if in span
64 × per (token, key)
ReLU(s·x) = s·ReLU(x), s > 0
(b) Blackwell (SM100): TMEM, packed BF16
TMEM
→ registers
cvt.rn.relu.bf16x2
ReLU rides on the convert
hfma2 × w
2 heads / instr
hadd2
reduce pairs
store BF16
if in span